A Quantity Theory of Money Approach to Cryptocurrency Valuation: Store of Value, Medium of Exchange, and the Economic Moat Signal

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Abstract

We apply the Quantity Theory of Money (QTM) to the cross-sectional valuation of cryptocurrencies and protocol tokens. Treating each digital asset as the currency of a self-contained economy, we decompose market capitalization into a Medium of Exchange (MoE) component, priced by the free-circulating velocity of supply, and a Store of Value (SoV) residual. The key innovation is λ—the fraction of total supply that is locked or staked—which partitions the circulating supply into economically active and inert components. A central result is that the SoV/MoE ratio simplifies to $\lambda/(1-\lambda)$, directly observable from on-chain data without velocity estimation. For native blockchain coins, high λ reflects a convenience yield and seigniorage signal grounded in monetary

search theory. For governance and utility tokens, which lack a medium-of-exchange function, λ itself is an economic moat signal, capturing token holder fundamental conviction in platform growth. We show that the conviction signal alone positively predicts future cross-sectional returns, and that this predictability is substantially stronger when the asset is cheap relative to its fundamental economic activity—measured by our Growth-Levelized NVT ratio for coins and the Growth-Levelized NV/TVL ratio for tokens—supporting the interpretation that valuation identifies when the conviction signal is not yet priced in. A long Stars/short Avoid portfolio generates positive factor-adjusted alpha after controlling for market, size, and momentum factors.

1 Introduction

What determines the value of a cryptocurrency? The dominant approach in the literature has been to adapt discounted cash flow (DCF) models, searching for a dividend-like flow—transaction fees, staking rewards, governance rights—that can be capitalized into a present value (Liu and Tsyvinski, 2021; Pagnotta and Buraschi, 2022; Biais, Bisière, Bouvard, and Casamatta, 2019). This approach, however, mischaracterizes the fundamental nature of digital assets. Cryptocurrencies and protocol tokens exhibit currency-like behavior: each coin or token is the monetary instrument of a self-contained digital economy, and its value is governed by the same forces that determine the value of any currency—the scale of economic activity it enables and the efficiency with which it circulates. The correct framework, we argue, is the Quantity Theory of Money (Fisher, 1911).

The Fisher equation, $MV = PQ$, relates the stock of money (M) and its velocity (V) to the nominal value of economic output (PQ) (Fisher, 1911). Applied to digital assets, a cryptocurrency's market capitalization proxies its money supply, and the economic output it supports—on-chain transaction volume for native coins, protocol throughput for tokens—is its PQ. This is not merely an analogy. Cong, Li, and Wang (2021) develop a formal dynamic model in which token prices aggregate heterogeneous users' transactional demand, with equilibrium prices governed by platform adoption—a formulation directly consistent with the QTM framework we develop here. Sockin and Xiong (2020) similarly model tokens as platform membership instruments, showing that the token price reflects the present value of the convenience yield accruing to participants: the non-pecuniary benefit of network membership, including access to platform services, network liquidity, and future transactional capability.

Before proceeding, we define the two components of value our framework decomposes. The *Medium of Exchange* (MoE) value of a digital asset is the value attributable to its current transactional function: the economic activity it enables today, priced by the velocity with which it circulates. The *Store of Value* (SoV) value is the premium agents are willing to pay today by forgoing current exchange utility—the transactions, goods, services, or alternative investments they could obtain by spending the asset now—in anticipation of future value appreciation. This appreciation is expected to arise from network growth, expanding user adoption, increasing utility, and innovation. An agent holding Ethereum rather than spending it is implicitly paying an intertemporal price: she sacrifices today's exchange utility for her conviction that the network's future worth exceeds today's use value. The SoV component is the aggregate market valuation of this forward-looking conviction.

The QTM framework requires a critical extension to operationalize this decomposition. Not all of a cryptocurrency's circulating supply is economically active. A substantial and growing fraction is locked in staking contracts, pledged as governance collateral, or held in long-term custody with near-zero velocity. These holdings are economically inert from a transactional standpoint: they contribute to the money supply but perform no current exchange function. Conflating active and inactive supply distorts velocity estimates and, consequently, misprices the economic content of market capitalization. We address this by introducing λ, defined as the proportion of the total supply that is locked or staked—including tokens committed to network validation, governance contracts, and other long-term holding functions. The formal derivations are provided in Appendix A. The key result is that the SoV/MoE ratio simplifies to

$$\frac{\mathrm{MC}_{\mathrm{SoV}}}{\mathrm{MC}_{\mathrm{MoE}}} = \frac{\lambda}{1-\lambda}, \tag{1}$$

a monotone, directly observable function of λ alone, requiring no velocity estimation.

Why does λ matter beyond its role in the decomposition? For native blockchain coins—Ethereum, Solana, and their peers—high λ is grounded in three complementary theoretical traditions. First, *monetary search theory* (Kiyotaki and Wright, 1993; Lagos and Wright, 2005) establishes that money emerges as a coordination equilibrium: an asset becomes valuable because rational agents expect others to hold and accept it. High λ is the blockchain-observable, publicly verifiable signal of this coordination commitment. Second, the *convenience yield* (Sockin and Xiong, 2020; Krishnamurthy and Vissing-Jorgensen, 2012) is the non-pecuniary benefit agents derive from holding a monetary asset: access to network security, platform liquidity, and future transactional capability. Just as holders of U.S. Treasury securities accept lower yields in exchange for liquidity services (Nagel, 2016), holders of dominant blockchain coins accept lower required returns because network membership has value beyond current transactions. Crucially, the SoV premium is the present value of expected *future* convenience yields: agents who lock coins today are paying for anticipated growth in the network's membership value. Third, the *seigniorage* channel (Cong and He, 2024) anchors this signal in rational optimization: in proof-of-stake networks, newly minted tokens distributed to stakers constitute monetary seigniorage, and the staking decision reveals belief that this income plus anticipated price appreciation exceeds the opportunity cost of locking capital. These three channels are complementary: coordination theory explains why high λ is self-reinforcing, convenience yield provides the holding motive, and seigniorage grounds the rational optimization.

Governance and utility tokens present a related but distinct case, and the distinction matters for how the framework is applied. A governance token such as UNI is the monetary instrument of the Uniswap protocol economy, but it has no medium-of-exchange function: the token does not need to change hands to facilitate Uniswap trades, so there is no MoE component to separate from an SoV premium. With the MoE denominator absent, the SoV/MoE ratio is undefined for tokens, and we use λ itself to convey the same conviction content directly. The protocol's economic output is still well defined—decentralized exchange volume, liquidity deployed, fees generated—and whether protocol revenue currently accrues to token holders is immaterial, just as it is immaterial whether U.S. GDP flows to dollar holders. This framing aligns with Cong, Li, and Wang (2021), who price tokens on the present value of future platform utility and adoption, and Schilling and Uhlig (2019), who model the value of digital currencies as functions of the economic activity they support. For governance tokens, λ carries a different signal. Staking a governance token into a DAO contract renders it illiquid: staked tokens cannot be sold while locked. An agent who voluntarily incurs this illiquidity cost reveals a preference for future appreciation over present liquidity—a choice grounded in assessments of network effects, utility expansion, competitive positioning, and innovation trajectory. Drawing on Hirschman (1970), holders who exercise *voice* (governance participation) rather than *exit* (selling) reveal a preference for the protocol's future that passive buyers do not. Edmans and Manso (2011) formalize this in a corporate setting: blockholders who engage rather than exit create governance value precisely because engagement is costly and therefore credible. Applied to tokens, active governance participation is a costly signal (Spence, 1973) of fundamental conviction in platform growth, making high participation rates an economic moat indicator.

We operationalize these ideas in two metrics per asset type. For coins, the *SoV/MoE ratio*—equal to $\lambda/(1-\lambda)$—measures the balance between forward-looking conviction and current transactional utility, and is our primary return predictor; for tokens, λ itself plays this role, as described above. The second metric normalizes market capitalization by fundamental economic activity, and its denominator likewise adapts to the asset type. For coins, the *Growth-Levelized NVT ratio* normalizes market capitalization by the present value of the network's expected transaction stream PQ (constructed analogously to a PMT-based discounted cash flow, as detailed in Section 3). For tokens, protocol throughput is measured by Total Value Locked rather than transaction volume, so the corresponding metric is the *Growth-Levelized NV/TVL ratio*: market capitalization over the growth-adjusted present value of TVL, built with the same discounting machinery as the coin metric. Crucially, neither valuation ratio carries its own independent theoretical prediction for returns. Rather, each serves as a conditioning variable: the conviction signal is most valuable as a return predictor when the market has not yet priced it in—that is, when the valuation ratio is low. Assets that combine high conviction with cheap valuation (Stars) offer the best expected returns; assets that combine low conviction with expensive valuation (Avoid) offer the worst.

We find that the conviction signal alone positively predicts future returns cross-sectionally for both coins and governance tokens. The predictive power is substantially stronger for assets with below-median valuation ratios, consistent with the view that the valuation ratio identifies when the conviction signal remains unpriced. A long Stars/short Avoid calendar-time portfolio generates an annualized alpha of [X]% with a Sharpe ratio of [Y] after controlling for market, size, and momentum factors, with the results holding separately for coins and governance tokens.

Related Literature. A growing body of work attempts to value digital assets, and our paper connects to several strands. On equilibrium cryptocurrency valuation, Pagnotta and Buraschi (2022) model Bitcoin value as a function of user adoption and network security; Biais, Bisière, Bouvard, and Casamatta (2019) study equilibrium selection in blockchain protocols; Schilling and Uhlig (2019) develop a general equilibrium model of Bitcoin as a competing currency; and Easley, O'Hara, and Basu (2019) analyze how transaction fees connect mining incentives to market value. Cong, Li, and Wang (2021) and Sockin and Xiong (2020) are the closest to our approach in modeling token prices through adoption dynamics and convenience yields respectively; we complement them by providing an empirically tractable QTM decomposition and a testable investment signal derived from on-chain observables. On crypto risk factors and return predictability, Liu and Tsyvinski (2021) identify network, momentum, and investor attention factors; Liu, Tsyvinski, and Wu (2022) develop a three-factor model for cryptocurrency returns; and Makarov and Schoar (2020) document price dislocations across exchanges. We contribute to this literature the first on-chain supply-side signal—$\lambda/(1-\lambda)$—grounded in monetary theory rather than return-based factor construction. On the monetary economics of digital assets, Kiyotaki and Wright (1993), Lagos and Wright (2005), and Schilling and Uhlig (2019) provide search-theoretic and general equilibrium foundations; Cong and He (2024) analyzes proof-of-stake tokenomics; and Jiang, Krishnamurthy, and Lustig (2024) draw the analogy between dollar exorbitant privilege and dominant-network premia. The convenience yield literature (Krishnamurthy and Vissing-Jorgensen, 2012; Nagel, 2016) motivates the monetary premium in $\mathrm{MC}_{\mathrm{SoV}}$. On token issuance and governance, Howell, Niessner, and Yermack (2020) study the determinants of ICO success; Chod and Lyandres (2021) model ICO design under information asymmetry; and Edmans and Manso (2011) and Hirschman (1970) provide the governance theory linking holder participation to firm and protocol value. Our paper is the first, to our knowledge, to derive a directly observable valuation decomposition from QTM, to distinguish the signal content of λ for coins versus governance tokens, and to demonstrate that the interaction of conviction and valuation predicts cross-sectional cryptocurrency returns.

The remainder of the paper is organized as follows. Section 2 develops the theoretical frameworks for coins (SoV/MoE) and tokens (governance moat) and derives the testable propositions. Section 3 defines and constructs the three key variables: the conviction index λ, the Growth-Levelized NVT ratio for coins, and the Growth-Levelized NV/TVL ratio for tokens. Section 4 describes the data sources, sample construction, and summary statistics. Section 5 presents the cross-sectional return predictability and portfolio results. Section 6 reports robustness tests. Section 7 concludes.

2 Theoretical Framework

This section develops the theoretical argument in three steps. Section 2.1 establishes the coin framework: QTM applied to native blockchain coins yields a SoV/MoE decomposition in which the locking ratio λ is the sufficient statistic. Section 2.2 develops the token framework: governance tokens lack a medium-of-exchange function, so λ is reinterpreted as a governance moat signal grounded in costly signaling theory. Section 2.3 introduces the common valuation-conditioning mechanism that applies across both tracks. Each subsection closes with the empirical propositions we test.

2.1 The Coin Framework: SoV/MoE and the Staking Ratio

The QTM identity applied to a native blockchain coin treats market capitalization as money supply: $\mathrm{MC} \cdot V = PQ$, where V is the velocity of supply in transactional use and PQ is nominal on-chain output. The critical observation is that supply is heterogeneous in velocity. A fraction λ of total circulating supply is locked in proof-of-stake validation contracts, staked to earn seigniorage, or otherwise immobilized with near-zero velocity. The complementary fraction $(1-\lambda)$ is freely circulating and economically active. Only the active fraction supports transactions; the locked fraction is held as a store of value and does not circulate. Partitioning supply accordingly, the appendix shows that market capitalization decomposes as

$$\mathrm{MC} = \underbrace{\frac{PQ}{V_{\text{active}}}}_{\mathrm{MC}_{\mathrm{MoE}}} + \underbrace{\frac{PQ}{V_{\text{active}}} \cdot \frac{\lambda}{1-\lambda}}_{\mathrm{MC}_{\mathrm{SoV}}},$$

and therefore the SoV/MoE ratio satisfies

$$\frac{\mathrm{MC}_{\mathrm{SoV}}}{\mathrm{MC}_{\mathrm{MoE}}} = \frac{\lambda}{1-\lambda}.$$

This is an accounting identity given λ, not yet a behavioral prediction.

The behavioral prediction follows from modeling the individual staking decision. Consider a continuum of agents indexed by $i \in [0,1]$, each choosing at date t whether to stake one unit of the native coin or hold it freely circulating. Staking yields a publicly observed current reward $b_t > 0$—seigniorage paid to validators in a PoS network (Cong and He, 2024)—plus a private convenience yield $c_{i,t+1} = \theta_{t+1} + \varepsilon_i$, where θ_{t+1} is the network's true future growth trajectory (unobserved at t) and ε_i is mean-zero idiosyncratic signal noise. Staking also imposes a private liquidity cost φ_i (i.i.d., independent of θ_{t+1} and ε_i). Agent i stakes if and only if

$$b_t + \theta_{t+1} + \varepsilon_i \geq \varphi_i \quad \Longleftrightarrow \quad \eta_i \leq b_t + \theta_{t+1}, \qquad \eta_i \equiv \varphi_i - \varepsilon_i.$$

By the law of large numbers, the aggregate staking ratio converges almost surely to

$$\lambda_t = H(b_t + \theta_{t+1}),$$

where H is the CDF of η_i. Idiosyncratic cost and signal noise wash out in the cross-section; what survives is a deterministic, strictly increasing function of θ_{t+1}. Because b_t is publicly observed, λ_t is a publicly verifiable signal of the holder base's aggregate private conviction about the network's future growth. Extracting θ_{t+1} from λ_t, however, requires knowing H and b_t with precision—not common knowledge for most market participants. As in Grossman and Stiglitz (1980), the signal is not fully impounded into price at t. As θ_{t+1} materializes and becomes public over $t+1$, price converges toward the value anticipated by high-λ stakers, generating positive return predictability.

Proposition 1a (Coins). $\mathrm{Cov}_j(r_{j,t+1}, \lambda_{j,t}/(1-\lambda_{j,t})) > 0$ in the cross-section: coins with higher SoV/MoE ratios earn higher subsequent returns, after controlling for size, momentum, and market beta.

Hypothesis 1a. *For native proof-of-stake coins, assets with higher $\lambda/(1-\lambda)$ in month t earn statistically and economically significant higher returns over month $t+1$, after controlling for coin-level size, momentum, and market exposure. The result holds regardless of whether the coin is an L1 or L2, provided the staking mechanism confers a positive seigniorage benefit $b_t > 0$.*

2.2 The Token Framework: Governance Conviction as Economic Moat

For governance and DeFi protocol tokens, the SoV/MoE framework is inapplicable: these tokens are not media of exchange. A holder of UNI does not need to spend UNI to use Uniswap; holding governance tokens does not reduce the protocol's transactional capacity. The token has no meaningful velocity in the QTM sense, and the SoV/MoE decomposition carries no economic content. What λ signals for tokens is something different: *governance conviction*—the fraction of supply whose holders have voluntarily incurred a cost to demonstrate fundamental belief in the protocol's future.

The formal model is a straightforward extension of the coin model. The same continuum of agents now hold a governance token at date t and choose whether to lock it into a governance contract (vote-escrow, delegation, or protocol staking) or hold it freely. Locking carries a private cost $\varphi_i = \ell + \kappa_i$, decomposed into a common illiquidity cost $\ell \geq 0$ (positive for vote-escrow protocols such as Curve's veCRV or Balancer's veBAL, zero for simple delegation under ERC20Votes) and an idiosyncratic engagement cost $\kappa_i \geq 0$, the monitoring and time cost of exercising governance voice rather than exit (Hirschman, 1970; Edmans and Manso, 2011). The current locking benefit $b_t \geq 0$ may include a fee or revenue share for protocols with on-chain distribution; for most governance tokens $b_t = 0$ and the incentive to lock is purely forward-looking. The threshold condition is identical to the coin case: $\eta_i = \varphi_i - \varepsilon_i \leq b_t + \theta_{t+1}$, yielding $\lambda_t = H(b_t + \theta_{t+1})$.

The key difference is interpretation. For coins, $b_t > 0$ is paid in newly minted seigniorage, so staking is partly a rational income decision. For tokens with $b_t = 0$, locking is a purely conviction-revealing action: the holder incurs $\ell + \kappa_i$ with no current compensation, revealing that their private belief about θ_{t+1} is strong enough to make the commitment worthwhile. This is a Spence (1973) costly-signaling equilibrium: only holders with sufficiently high conviction rationally pay the cost. The resulting aggregate λ_t is therefore a cleaner signal of fundamental conviction for governance tokens than for coins, because the seigniorage confound is absent. The partial-revelation logic applies identically: the signal in λ_t is not freely extractable from public data, so it is not instantly priced, generating the same return predictability.

Proposition 1b (Tokens). $\mathrm{Cov}_j(r_{j,t+1}, \lambda_{j,t}) > 0$ in the cross-section of governance tokens: assets with higher governance conviction ratios earn higher subsequent returns, after controlling for size, momentum, and protocol category.

Hypothesis 1b. *For DeFi governance tokens, assets with higher λ in month t earn statistically and economically significant higher returns over month $t+1$, after controlling for token-level size, momentum, and protocol category. The predictive power of voting-participation-weighted λ exceeds that of passive locking alone, consistent with the costly-signaling logic: the harder the action, the purer the signal.*

2.3 The Valuation Conditioning Signal

Propositions 1a and 1b establish that λ carries information not yet fully in price. What they do not characterize is how much of the signal remains unpriced at a given date. Let $\delta_t \in [0, 1]$ denote the fraction of the conviction signal in λ_t already absorbed into the current price, so that $p_t = p_t^{\text{fund}}(1 + \delta_t)$ where p_t^{fund} reflects current economic activity. The expected return as the signal is validated over $t+1$ is proportional to the unabsorbed portion $\lambda_t(1 - \delta_t)$, not to λ_t alone. The marginal predictive power of λ_t is therefore decreasing in δ_t, and δ_t is increasing in the current market capitalization relative to fundamental activity.

For coins, the relevant fundamental is discounted expected on-chain volume: δ_t is increasing in $\text{NVT\_GL}_t = \text{MC}_t/PQ_t^*$, where PQ_t^* is the growth-adjusted present value of expected transaction activity (formally defined in Section 3.2). For tokens, the relevant fundamental is the growth-adjusted present value of Total Value Locked: δ_t is increasing in $\text{NV/TVL\_GL}_t = \text{MC}_t/\text{TVL}_t^*$, where TVL_t^* is constructed with the same discounting machinery (Section 3.3). In both cases, a high ratio indicates that current market capitalization already embeds the conviction signal, and the incremental return predictability of λ_t is low; a low ratio indicates the signal remains unabsorbed.

Proposition 2. *The cross-partial effect of λ_t and the valuation ratio on expected returns is negative: the return predictability of λ_t is concentrated among low-valuation-ratio assets.*

Hypothesis 2. *The return predictability documented in H1a and H1b is significantly stronger among assets with below-median valuation ratios (NVT_GL for coins, NV/TVL_GL for tokens). The marginal return to a unit increase in λ is higher when the asset is cheap* *relative to its fundamental economic activity.*

The valuation ratio carries no independent return prediction in this framework. A low NVT_GL alone—without a high λ—merely describes a coin with limited current throughput relative to market cap, not necessarily a mispricing; similarly for NV/TVL_GL. The conditioning role is asymmetric: valuation identifies the *timing* of conviction signal absorption, not a separate source of alpha.

2.4 Hypotheses and the Quadrant Portfolio

Propositions 1 and 2 jointly define a two-dimensional framework. *Stars* combine high λ with low valuation ratio—strong conviction, signal not yet priced—and are predicted to have the highest future returns. *Avoid* assets combine low λ with high valuation ratio—weak conviction, valuation already stretched—and are predicted to have the lowest returns.

Hypothesis 3. *A long-short portfolio that goes long Star assets (above-median λ, below-median valuation ratio) and short Avoid assets (below-median λ, above-median valuation ratio), rebalanced monthly, generates positive and statistically significant factor-adjusted alpha after controlling for market, size, and momentum. The result holds separately within the coin sample (using NVT_GL) and within the token sample (using NV/TVL_GL), and in the combined panel. The Stars-minus-Avoid Sharpe ratio exceeds that of equal-weighted benchmark portfolios.*

H3 is the integrating test: it asks whether the joint two-dimensional signal has incremental predictive power beyond either dimension alone. We report results separately for coins and tokens to preserve the track distinction: for coins, alpha reflects the return to SoV/MoE mispricing correcting over time; for tokens, alpha reflects the market's underreaction to governance conviction signals.

3 Variable Construction

The theoretical framework in Section 2 requires three empirical proxies: the conviction index λ, common to both asset tracks; the Growth-Levelized NVT ratio for the coin track; and the Growth-Levelized NV/TVL ratio for the token track. This section defines each variable and describes its construction. Data sources and sample formation are covered in Section 4.

3.1 The Conviction Index λ

The theoretical λ_t is the aggregate locking or commitment ratio of an asset's holder base at date t. Empirically, the conviction-revealing actions observable on-chain vary by asset type and protocol design. We therefore construct λ as a composite index over three conceptually motivated channels, and aggregate across available channels using cross-sectional z-scores.

Channel 1: Staking. For proof-of-stake layer-one coins, the staking channel is the fraction of circulating supply bonded as validator or delegator stake at month-end. This is the direct empirical counterpart of the locking action in Section 2.1: it simultaneously earns seigniorage ($b_t > 0$) and constitutes a costly, publicly visible commitment to the network's future. Staking rates are sourced from chain-native archive data for each coin.[1] The staking channel covers 48 assets and 2,222 asset-months.

Channel 2: Holding. The holding channel measures the fraction of circulating supply that has not transacted on-chain in the preceding six months (HODL-6m). This proxy extends conviction measurement to any EVM-compatible asset without a formal staking mechanism, mapping to the inert supply fraction λ_t in the model: a holder who has not

[1] For Polkadot (DOT) and Kusama (KSM), bonded supply is retrieved from the Subscan archive API. For Cosmos-ecosystem chains (CRO, KAVA, OSMO), we query the Cosmos SDK Staking module endpoint at block heights corresponding to month-end timestamps, with timestamps identified via binary search on the CometBFT RPC interface. For EVM-based staking contracts (BNB, MATIC, TRX, and others), staked supply is derived from on-chain staking event logs via the Etherscan multi-chain archive API. Each data source is validated by confirming that reported values vary across block heights, ensuring the archive reflects historical rather than current-only state.

transacted for six months has revealed a preference for holding over liquidity. HODL-6m is computed by replaying on-chain Transfer event logs in chronological order under a FIFO attribution rule, assigning each unit of supply to its most recent transfer date.[2] This channel spans 408 assets and 11,654 asset-months, providing the broadest coverage in the conviction panel.

Channel 3: Governance. Two governance sub-channels capture active participation in protocol decision-making. The delegation sub-channel records the fraction of circulating supply delegated to a voting representative under an ERC20Votes contract, observable from on-chain Delegate event logs.[3] The voting sub-channel records proposal participation rates from Snapshot off-chain governance records, cross-referenced with on-chain ERC20Votes logs where available. Both sub-channels identify holders who have incurred the engagement costs κ_i of Section 2.2—the costlier the action, the more selected and informative the resulting signal. The delegation and voting sub-channels cover 26 and 53 assets, respectively, concentrated in the DeFi governance token sample.

Composite Index. For each asset-month, each available channel is standardized to a z-score over all asset-months with non-missing observations for that channel. The composite λ_z is the equal-weight average of standardized channel scores available for that asset-month. The result has approximately zero cross-sectional mean and is defined whenever at least one channel is observed. Equal weighting imposes no prior on the relative information content of the channels; channel-specific predictive power is examined in robustness tests. The full conviction panel covers 467 assets and 13,626 asset-months, of which 11,901 are

[2]Transfer events are retrieved from the Etherscan Pro V2 multi-chain archive, covering Ethereum mainnet and twelve additional EVM chains. Two data quality filters are applied: (i) monthly on-chain flow exceeding 100× reported circulating supply is classified as a minting, bridging, or aggregator artifact and excluded; (ii) asset-months in which HODL-6m exceeds 80% persistently are flagged for robustness checks, consistent with possible custodial cold-wallet concentration rather than individual holder conviction. Self-transfers (sender equals recipient) are excluded from the event replay.

[3]Data sourced from `DelegateVotesChanged` event logs via the Etherscan Pro V2 API, aggregated to month-end totals.

single-channel, 1,371 two-channel, and 354 three-or-more-channel observations.

3.2 Growth-Levelized NVT

For the coin regression track, the valuation anchor is the Growth-Levelized Network Value to Transactions ratio (NVT_GL), which scales market capitalization by the growth-adjusted, risk-discounted present value of expected on-chain throughput:

$$\text{NVT\_GL}_{j,t} = \frac{\text{MC}_{j,t}}{PQ^*_{j,t}}, \tag{2}$$

$$PQ^*_{j,t} = \frac{\displaystyle\sum_{s=1}^{n} \frac{PQ_{0,j,t}(1+g_{j,t})^s}{(1+r_{e,j,t})^s} + \frac{PQ_{0,j,t}(1+g_{j,t})^n(1+g_\infty)}{(r_{e,j,t}-g_\infty)(1+r_{e,j,t})^n}}{\dfrac{1-(1+r_{e,j,t})^{-n}}{r_{e,j,t}}}, \tag{3}$$

where $PQ_{0,j,t}$ is the trailing twelve-month annualized on-chain transfer volume for asset j at month t; $g_{j,t}$ is the trailing three-year compound annual growth rate of PQ_0, capped in $[-50\%, 200\%]$; $n = 10$ years is the explicit-growth projection horizon; $g_\infty = 3\%$ is the terminal perpetuity growth rate; and $r_{e,j,t} = r_f + \hat{\beta}_{j,t} \cdot \text{MRP}$ is the asset-specific discount rate, with $r_f = 4\%$, $\text{MRP} = 30\%$, and $\hat{\beta}_{j,t}$ the trailing 36-month beta against the BTC price index, floored so that $r_{e,j,t} \geq g_\infty + 2\%$ to ensure well-defined terminal values.[4]

NVT_GL below one signals that market capitalization is below the growth-adjusted present value of expected on-chain activity. NVT_GL substantially above one indicates that current market capitalization already embeds forward growth not yet visible in throughput. The growth adjustment corrects a systematic bias in raw NVT: fast-growing assets may appear expensive on a raw NVT basis but inexpensive once expected growth is discounted.

[4] The denominator of (3) is the standard annuity factor normalizing the finite-horizon present value of future PQ to an equivalent annualized flow, making NVT_GL interpretable as a price-to-normalized-activity ratio analogous to a price-to-normalized-earnings multiple in equity analysis. The MRP of 30% reflects the elevated required return on cryptocurrency risk; $\hat{\beta}_{j,t}$ is retained in the panel so alternative discount rate assumptions can be applied directly in robustness tests.

3.3 Growth-Levelized NV/TVL

For the token regression track, the valuation anchor is the Growth-Levelized Network Value to Total Value Locked ratio, constructed with the same discounted-present-value machinery as NVT_GL. Total Value Locked—the market value of assets deposited in a protocol's smart contracts—proxies for the scale of economic activity the protocol intermediates, and is the token analog of on-chain transaction volume in the QTM framework. Because TVL is a stock rather than a flow, the base is a trailing average level rather than a trailing sum:

$$\text{NV/TVL\_GL}_{j,t} = \frac{\text{MC}_{j,t}}{\text{TVL}^*_{j,t}}, \tag{4}$$

$$\text{TVL}^*_{j,t} = \frac{\displaystyle\sum_{s=1}^{n} \frac{\text{TVL}_{0,j,t}(1+g_{j,t})^s}{(1+r_{e,j,t})^s} + \frac{\text{TVL}_{0,j,t}(1+g_{j,t})^n(1+g_\infty)}{(r_{e,j,t}-g_\infty)(1+r_{e,j,t})^n}}{\dfrac{1-(1+r_{e,j,t})^{-n}}{r_{e,j,t}}}, \tag{5}$$

where $\text{TVL}_{0,j,t}$ is the trailing twelve-month average of month-end TVL for protocol j (requiring at least six monthly observations); $g_{j,t}$ is the trailing three-year compound annual growth rate of TVL_0, with two- and one-year fallbacks and the same [−50%, 200%] cap; and all remaining parameters—n, g_∞, r_f, MRP, and the beta construction—are identical to those in (3), so the coin and token tracks share a single set of valuation assumptions. Where a protocol is deployed across multiple chains, TVL is the sum of all-chain TVL.

The growth adjustment serves the same purpose as in the coin track: a protocol whose TVL is expanding rapidly may appear expensive on a raw NV/TVL basis but inexpensive once expected growth is priced in, and conversely a protocol with collapsing TVL may appear deceptively cheap on the raw ratio. A low NV/TVL_GL indicates that market capitalization is modest relative to the growth-adjusted present value of the protocol's economic footprint; a high NV/TVL_GL indicates the reverse.

4 Data

4.1 Data Sources

Price, market capitalization, and circulating supply are drawn from CoinMarketCap at daily granularity, snapped to end-of-calendar-month observations. CoinMarketCap provides the broadest historical coverage of digital asset market data and is the standard reference for universe construction in the cryptocurrency finance literature (Liu, Tsyvinski, and Wu, 2022; Makarov and Schoar, 2020).

On-chain Transfer event logs, staking event logs, and ERC20Votes Delegate event logs are retrieved from the Etherscan Pro V2 multi-chain API, which provides archive access to EVM-compatible chains including Ethereum mainnet, BNB Chain, Polygon, Avalanche, Arbitrum, Optimism, and nine additional networks. Non-EVM staking data follows chain-native sources: Subscan archive API for Polkadot and Kusama; Cosmos SDK Staking module endpoints for Cosmos-ecosystem chains. Off-chain governance participation is sourced from the Snapshot API, cross-referenced with on-chain ERC20Votes logs.

On-chain transaction volume for the NVT_GL construction is assembled from three sources in priority order. DeFiLlama chain-level DEX volume[5] serves as the primary source for chains with active decentralized exchange infrastructure (1,340 asset-months). BitInfoCharts sent-in-USD settlement value[6] is used for PoW coins and pre-DEX periods, capturing all on-chain transfers rather than DEX activity alone (705 asset-months). Blockchain.com estimated transaction value[7] supplements Bitcoin and legacy chain observations with pre-DEX-era coverage (118 asset-months). The remaining 363 asset-months use DeFiLlama protocol-level volume for assets whose primary activity flows through specific protocols. Each series is validated against the chain's own reported aggregate, with the reconstructed monthly level required to lie within 5% of the chain total. The resulting NVT_GL panel

[5] `https://defillama.com`, accessed through May 2026.
[6] `https://bitinfocharts.com`, accessed through May 2026.
[7] `https://blockchain.com`, accessed through May 2026.

covers 67 assets and 2,526 valid asset-months from August 2016 through May 2026.

Total Value Locked is sourced from DeFiLlama at daily granularity, snapped to month-end; multi-chain protocols use the sum of all-chain TVL. The raw TVL panel covers 162 assets and 8,066 asset-months from September 2017 through May 2026. Applying the growth-levelization of (5)—which requires a trailing twelve-month TVL average and at least one year of prior history for the growth estimate—yields an NV/TVL_GL panel of 111 assets and 3,125 asset-months from January 2021 through May 2026.

4.2 Sample Construction

The starting universe is 1,939 digital assets drawn from the CoinMarketCap historical database, observed at end-of-calendar-month from August 2015 through May 2026, yielding 156,838 asset-month observations.[8] Table 1 traces the path from this universe to the two regression samples; we walk through each step in turn.

The first cut is by asset class. Each of the 1,939 assets is classified as a *coin* (native settlement asset of a public blockchain, e.g., BTC, ETH, SOL), a *token* (smart-contract-issued asset deployed on a host chain, e.g., ERC-20 governance and DeFi tokens), or *other*. The other class—858 assets comprising stablecoins, wrapped and bridged representations, and exchange or infrastructure assets whose value is structurally indexed to an external reference—is excluded outright: neither the SoV/MoE framework nor the governance moat framework applies to an asset whose price is pegged or mechanically derived. This leaves 1,081 candidates: 633 coins and 448 tokens.

The coin track requires a staking channel through which conviction is expressed and seigniorage earned (Section 2.1). Of the 633 coins, 323 are proof-of-work only and have no staking mechanism; they are outside the scope of Hypothesis 1a, though they contribute to NVT_GL baseline tests. A further 24 could not be verified as proof-of-stake from chain

[8] All assets with at least one non-zero price observation in CoinMarketCap are included. Market capitalization is computed as reported price times circulating supply. Continuous coverage is not required; assets may enter and exit the panel as they gain or lose listing.

documentation. Of the remaining 286 proof-of-stake candidates, historical staking ratios could be reconstructed for 30;[9] of these, 24 also have the on-chain volume history required for NVT_GL. The *coin regression sample* is therefore 24 assets and 718 asset-months spanning December 2020 through May 2026, with median NVT_GL of 0.017 and median market capitalization of $820 million.

The token track requires TVL as the fundamental denominator. Of the 448 tokens, 299 have no DeFiLlama TVL coverage—predominantly utility, memetic, and infrastructure tokens for which protocol TVL is not defined—leaving 149. Three of these lack any conviction channel observation, leaving 146 with both λ_z and TVL. Of these, 45 drop out of the growth-levelized sample because their TVL history is too short to estimate a growth rate or does not overlap with their conviction observations.[10] The *token regression sample* is therefore 101 assets and 2,771 asset-months spanning January 2021 through May 2026, with median NV/TVL_GL of 2.29 (IQR: 0.20–16.10), median TVL of $42.2 million, and median market capitalization of $98 million.

The combined regression panel spans 125 unique assets and 3,489 asset-months. At the end of the sample in May 2026, the panel accounts for $451 billion in market capitalization—19.0% of the $2.38 trillion CoinMarketCap universe—consistent with the concentration of on-chain governance and staking infrastructure among the larger assets in the market. Channel 2 (holding) applies only to EVM-compatible assets; PoW coins with UTXO-based ledger accounting require a separate pipeline based on UTXO age analysis and are left to future work.

4.3 Summary Statistics

Table 2 reports coverage across data panels and key statistics for the regression samples.

[9] The binding constraint is archive depth: most chains' public endpoints serve only recent state, either because nodes prune aggressively or because historical queries are routed through commercial indexers. Chains with recoverable history are documented in Section 4.1.

[10] The growth-levelization requirement shortens the token sample relative to a raw NV/TVL alternative (136 assets, 4,627 asset-months from March 2020): assets without at least one year of prior TVL history

Table 1: Sample Construction

	Excluded	Remaining
CoinMarketCap universe, 2015-08 to 2026-05		1,939
Less: other class (stablecoins, wrapped, infrastructure)	858	1,081
Coin track		
Coins		633
Less: proof-of-work only (no staking channel)	323	310
Less: consensus type unverifiable	24	286
Less: historical staking data unrecoverable	256	30
Less: no on-chain volume overlap (NVT_GL)	6	**24**
Token track		
Tokens		448
Less: no DeFiLlama TVL coverage	299	149
Less: no conviction channel observed	3	146
Less: TVL history insufficient for growth estimate	45	**101**
Combined regression panel		**125**

Notes: Each row removes assets failing the stated requirement, conditional on surviving all prior rows. The coin regression sample requires proof-of-stake architecture, at least one month of reconstructed staking-based λ_z, and overlapping positive NVT_GL. The token regression sample requires DeFiLlama TVL coverage, at least one conviction channel, and sufficient TVL history for the growth-levelized denominator TVL* (at least one year).

5 Results

[To be written.]

6 Robustness

[To be written.]

7 Conclusion

[To be written.]

have no growth estimate and drop out. The raw NV/TVL sample is retained for robustness comparisons.

Table 2: Summary Statistics

Panel / Sample	Assets	Asset-months	Date range	Median MC
Full universe	1,939	156,838	2015-08 to 2026-05	\$6.5M
Conviction panel (λ_z)	467	13,626	2016-12 to 2026-05	—
ch1 staking	48	2,222	2017-07 to 2026-05	—
ch2 holding	408	11,654	2015-08 to 2026-05	—
ch3 delegation	26	627	2020-05 to 2026-05	—
ch3 voting	53	1,202	2020-08 to 2026-05	—
NVT_GL panel	67	2,526	2016-08 to 2026-05	—
TVL panel (raw)	162	8,066	2017-09 to 2026-05	—
NV/TVL_GL panel	111	3,125	2021-01 to 2026-05	—
Coin regression (NVT_GL)	24	718	2020-12 to 2026-05	\$820M
Token regression (NV/TVL_GL)	101	2,771	2021-01 to 2026-05	\$98M
Combined	125	3,489	2020-12 to 2026-05	\$138M

Notes: Median MC is the cross-sectional median market capitalization across all asset-months with a positive price observation. Conviction panel sub-channel counts sum to more than the panel total because 1,725 asset-months appear in two or more channels. Coin regression requires simultaneous non-missing λ_z and positive NVT_GL for proof-of-stake coins. Token regression requires simultaneous non-missing λ_z and positive NV/TVL_GL. Median NVT_GL in the coin regression sample is 0.017. Median NV/TVL_GL in the token regression sample is 2.29 (IQR: 0.20–16.10); median TVL is \$42.2M.

A Derivation of the SoV/MoE Decomposition

This appendix provides the formal derivation of the SoV/MoE decomposition and its simplification to a function of λ alone.

Setup. Let M denote the total circulating supply of a digital asset and MC its market capitalization, which serves as the money supply proxy in the QTM framework. Let PQ denote nominal economic output (on-chain transaction volume for coins; protocol throughput for tokens). Standard QTM gives $\text{MC} \cdot V = PQ$, so the market capitalization satisfies $\text{MC} = PQ/V$.

The locking ratio. Define λ as the proportion of total supply that is locked or staked:

$$\lambda = \frac{M_{\text{locked}}}{M}, \qquad \lambda \in [0, 1).$$

The free-circulating supply is $M(1-\lambda)$. We define the *free velocity* V_{free} as the velocity of only the economically active (unlocked) portion:

$$V_{\text{free}} = \frac{PQ}{M(1-\lambda)}.$$

MoE and SoV components. The Medium of Exchange component of market capitalization is the value attributable to current transactional activity, priced at the free velocity:

$$\text{MC}_{\text{MoE}} = \frac{PQ}{V_{\text{free}}} = M(1-\lambda).$$

The Store of Value component is the residual:

$$\text{MC}_{\text{SoV}} = \text{MC} - \text{MC}_{\text{MoE}} = M - M(1-\lambda) = M\lambda.$$

The SoV/MoE ratio. The ratio of the SoV to MoE components is:

$$\frac{\text{MC}_{\text{SoV}}}{\text{MC}_{\text{MoE}}} = \frac{M\lambda}{M(1-\lambda)} = \frac{\lambda}{1-\lambda}.$$

This result shows that the SoV/MoE ratio is a monotone increasing function of λ alone, requiring no estimation of velocity, transaction volume, or any other derived quantity. As $\lambda \to 0$, the ratio approaches zero (pure medium of exchange). As $\lambda \to 1$, the ratio diverges (pure store of value).

The Growth-Levelized NVT ratio. The Network Value to Transactions ratio is NVT $=$ MC$/PQ_{\text{annualized}}$. We construct a forward-looking version by replacing current PQ with PQ^*, the levelized present value of the projected transaction stream:

$$PQ^* = \frac{\sum_{s=1}^{n} \frac{PQ_0(1+g)^s}{(1+r_e)^s} + \frac{PQ_0(1+g)^n(1+g_\infty)}{(r_e-g_\infty)(1+r_e)^n}}{\text{annuity factor}(r_e, n)},$$

where g is the trailing k-year CAGR of PQ, r_e is the CAPM-estimated required return, g_∞ is the terminal growth rate, and the annuity factor converts the present value to a levelized annual equivalent (analogous to the Excel PMT function). The Growth-Levelized NVT ratio is then $\text{NVT}_{GL} = \text{MC}/PQ^*$. Full estimation details are in Section 4.

References

BitInfoCharts, 2026, Blockchain statistics and sent-value data, `https://bitinfocharts.com`, accessed May 2026.

Blockchain.com, 2026, Bitcoin blockchain explorer and transaction statistics, `https://blockchain.com`, accessed May 2026.

Biais, B., C. Bisière, M. Bouvard, and C. Casamatta, 2019, The blockchain folk theorem, *Review of Economic Studies* 86, 1341–1371.

CoinMarketCap, 2026, Historical cryptocurrency market data, `https://coinmarketcap.com`, accessed May 2026.

Chod, J., and E. Lyandres, 2021, A theory of ICOs: Diversification, agency, and information asymmetry, *Management Science* 67, 5969–5989.

Cong, L. W., Y. Li, and N. Wang, 2021, Tokenomics: Dynamic adoption and valuation, *Review of Financial Studies* 34, 1105–1155.

Cong, L. W., and Z. He, 2024, The tokenomics of staking, *NBER Working Paper* 33640.

DeFiLlama, 2026, Protocol TVL and decentralized exchange volume data, `https://defillama.com`, accessed May 2026.

Easley, D., M. O'Hara, and S. Basu, 2019, From mining to markets: The evolution of Bitcoin transaction fees, *Journal of Financial Economics* 134, 91–109.

Edmans, A., and G. Manso, 2011, Governance through exit and voice: A theory of multiple blockholders, *Review of Financial Studies* 24, 2395–2428.

Fisher, I., 1911, *The Purchasing Power of Money*, Macmillan, New York.

Grossman, S. J., and J. E. Stiglitz, 1980, On the impossibility of informationally efficient markets, *American Economic Review* 70, 393–408.

Hirschman, A. O., 1970, *Exit, Voice, and Loyalty*, Harvard University Press, Cambridge, MA.

Howell, S. T., M. Niessner, and D. Yermack, 2020, Initial coin offerings: Financing growth with cryptocurrency token sales, *Review of Financial Studies* 33, 3925–3974.

Jiang, Z., A. Krishnamurthy, and H. Lustig, 2024, Exorbitant privilege: A safe-asset view, *NBER Working Paper* 32454.

Kiyotaki, N., and R. Wright, 1993, A search-theoretic approach to monetary economics, *American Economic Review* 83, 63–77.

Krishnamurthy, A., and A. Vissing-Jorgensen, 2012, The aggregate demand for Treasury debt, *Journal of Political Economy* 120, 233–267.

Lagos, R., and R. Wright, 2005, A unified framework for monetary theory and policy analysis, *Journal of Political Economy* 113, 463–484.

Liu, Y., and A. Tsyvinski, 2021, Risks and returns of cryptocurrency, *Review of Financial Studies* 34, 2689–2727.

Liu, Y., A. Tsyvinski, and X. Wu, 2022, Common risk factors in cryptocurrency, *Journal of Finance* 77, 1133–1177.

Makarov, I., and A. Schoar, 2020, Trading and arbitrage in cryptocurrency markets, *Journal of Financial Economics* 135, 293–319.

Nagel, S., 2016, The liquidity premium of near-money assets, *Quarterly Journal of Economics* 131, 1927–1971.

Pagnotta, E., and A. Buraschi, 2022, An equilibrium valuation of Bitcoin and decentralized network assets, *Journal of Finance* 77, 887–943.

Schilling, L., and H. Uhlig, 2019, Some simple Bitcoin economics, *Journal of Monetary Economics* 106, 16–26.

Sockin, M., and W. Xiong, 2020, A model of cryptocurrencies, *NBER Working Paper* 26816.

Spence, M., 1973, Job market signaling, *Quarterly Journal of Economics* 87, 355–374.